

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	CSM-B9
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Context-Aware DevOps Root Cause Analysis Using Hybrid RAG, Transformers, and LLMs

Abstract

DevOps and cloud-native environments generate large volumes of heterogeneous operational data, including logs, deployment events, metrics, traces, configuration changes, and version-control information. Although recent approaches use Machine Learning, Large Language Models (LLMs), and probabilistic reasoning for anomaly detection and root cause analysis, incident diagnosis remains challenging because relevant evidence is distributed across multiple sources, historical incidents are often underutilized, and LLM-generated diagnoses may lack systematic evidence selection. This project proposes a **Context-Aware Hybrid Retrieval-Augmented Generation (RAG) framework** for explainable DevOps incident root cause analysis. The research is motivated by recent work on cloud-native anomaly detection and root cause analysis using LLMs and Bayesian Networks, as well as research highlighting the importance of observability in cloud-native applications. The proposed framework integrates **semantic retrieval using BAAI/bge-small-en-v1.5 and lexical retrieval using BM25** to retrieve relevant technical documentation, historical incidents, logs, deployment information, and project-specific evidence. A **BGE reranker** prioritizes the retrieved evidence according to the current incident context. A **fine-tuned DeBERTa-v3-base Transformer** then analyzes the selected evidence and ranks candidate root causes with confidence scores. Finally, **Qwen2.5-7B-Instruct** generates an evidence-grounded explanation and recommends appropriate remediation steps. The system will be implemented as a full-stack AI DevOps platform using **React, FastAPI, PostgreSQL with pgvector, Docker, Kubernetes, Jenkins, GitHub, and Prometheus**, enabling automated collection and analysis of DevOps telemetry. The framework will be evaluated using **RCAEval** together with controlled domain-specific incidents involving Kubernetes, Docker, Jenkins, Git, and application failures. Experiments will compare conventional retrieval, LLM-based diagnosis, and the proposed Hybrid RAG approach using root-cause ranking accuracy, Precision, Recall, F1-Score, evidence relevance, response time, and explanation quality. Ablation experiments will further examine the individual contribution of hybrid retrieval, reranking, and Transformer-based root-cause ranking. The expected outcome is an explainable AI-assisted DevOps incident management system that provides traceable relationships between retrieved evidence, predicted root causes, and generated explanations, thereby improving the reliability and effectiveness of automated root cause analysis.

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References (Category specific)

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